

Aswin VATTAPPARAMBATHU JAYAPRAKASH

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EDUCATION

IMLEX - MSc Engineering, Lighting and XR | MSc in Optics, Image, Computer Vision, Machine Learning and Multimedia | MSc of Computer Science • Sep 2024 - Sep 2026

University of Eastern Finland | Universite Jean-Monnet | Toyohashi University of Technology • Finland | France | Japan

- Erasmus Mundus Joint Master's degree in Imaging and Light in Extended Reality, studying across three universities in Finland, France, and Japan
- Multidisciplinary curriculum combining photonics, computer vision, robotics, and XR technologies, including courses in eye tracking, deep learning, data science, and human perception with industry collaboration
- Master's Thesis (Upcoming): "Industrial Training Using VR/AR-Based Prototyping and AI" at National Institute of Advanced Industrial Science and Technology (AIST), Japan (Mar–Aug 2026)

BSc (honors) in Physics • Hindu College, University of Delhi • Delhi, India Nov 2020 - Jul 2023

EXPERIENCE

XR Research and Development Intern

LIRIS (Laboratoire d'InfoRmatique en Image et Systèmes d'information) • Saint-Étienne, France Jun 2025 - Sep 2025

- Enhanced open-source 6DoF VR Thermal Feedback System (post-IEEE VR 2025 publication) by implementing obstacle-aware thermal rendering using Unity 3D raycasting and collision detection in C#; merged contributions to main repository, improving system realism for future XR haptics research
- Integrated a haptic vest with custom vibrotactile feedback patterns synchronized to gameplay events
- Developed multisensory VR demo combining thermal lamps, haptic vest, and spatialized audio (via ASIO Audio Manager plugin) in a road-crossing scenario; user testing confirmed enhanced immersion compared to baseline VR-only experience
- Improved developer experience by adding intuitive material system (color-coded thermal sources), teleport navigation, and hit-feedback visualization to demo scenes

PROJECTS

VR Perception Study with Physical Props | Unity, XR Interaction Toolkit, OpenXR Nov 2025 - Nov 2025

Toyohashi University of Technology, Japan

Link: https://aswin-avj.github.io/portfolio/projects_page/PlankExp/PlankExp.html

- Developed Unity-based experimental framework with a "walk on the plank" scenario for XR perception research investigating how physical realism influences height perception and fear responses in VR
- Engineered randomized trial system with elevator mechanics and height-adjustable plank scenarios
- Implemented automated data logging for subjective height judgments, movement timing, and behavioral metrics; results showed physical plank lowered Point of Subjective Equality (PSE) and increased cautious movement time

AI-Based Lighting Transfer Using Synthetic Data | PyTorch, Blender, Git Mar 2025 - May 2025

Jean Monnet University, France

Link: https://aswin-avj.github.io/portfolio/projects_page/f2b.html

- Synthesized photorealistic dataset of 1,600 paired samples (16 face models × 100 lighting/viewpoint conditions) using 3D reconstruction tools (KeenTools FaceBuilder (in Blender), TRELIS AI) and consistent rendering pipeline
- Designed and trained U-Net architecture with ResNet34 encoder for facial relighting tasks; implemented hybrid loss function (MSE + SSIM) achieving 15% SSIM improvement over 500 training epochs
- Validated end-to-end pipeline demonstrating successful lighting transfer (shadows, highlights, light direction) from facial images to reference sphere

Real-Time 3D Color Cloud Visualization | WebGL, Three.js, Shader Programming Feb 2025 - Feb 2025

Jean Monnet University, France

Link: https://aswin-avj.github.io/portfolio/projects_page/color_space.html

- Designed and implemented custom GLSL shaders for rendering RGB, CIExyY, and CIELAB color spaces in 3D point clouds
- Developed VR/MR interactions allowing users to manipulate video feeds and color clouds with 6DoF controllers
- Built real-time processing pipeline handling video input and color space transformations at 60fps

***Additional projects and professional experience available at:**

aswin-avj.github.io/portfolio

SKILLS

Programming languages & Frameworks: Python, C++, Unity (C#), C#, WebXR, Three.js, PyTorch, Keras, ROS 2, SLAM, Scilab (Matlab alternative)

XR/Graphics: OpenXR, XR Interaction Toolkit, Meta Quest SDK, GLSL Shader Programming, WebGL, Real-time 3D Visualization

Tools: Git/GitHub, CUDA, DOTween, LaTeX, DaVinci Resolve

Languages: English (Bilingual), Hindi (Bilingual), Malayalam (Native), French (Beginner), Japanese (Beginner)

VOLUNTEERING & LEADERSHIP

Panchtatva, Environment Society Jan 2022 - Jan 2023
Media Head • Delhi

- Managed video production and graphic design for sustainability campaigns reaching 10,000+ audience members

Eureka, Physics Society Jan 2022 - Jan 2023
Head Member • Delhi

- Led video creation for AI and astronomy workshops; co-organized scientific literacy events